Project Report: TripAdvisor Recommendation System

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1. The Idea

The main goal of this project was to build a system that recommends the most similar place based on the user reviews. Basically, if I love a specific restaurant, the system should find me another one just like it.

To do this, we started with a assumption: similar experiences (whether it's a hotel, restaurant, or attraction) are expressed in a similar way through text. If this is true, we should be able to make great recommendations just by reading what clients say, without even looking at the official metadata of the place.

2. Data Preparation

Before throwing complex algorithms at our data, we had to clean things up. First, to simplify the problem, we filtered the dataset to keep only the English reviews.

But we ran into a issue: the number of reviews per place is extremely variable. Some touristy spots have thousands of comments, while others have barely two. To reduce this differences in size, we applied a filtering strategy as suggested in the guidelines : we decided to limit the number of reviews, keeping only the first 10 for each place. This perfectly balanced our data and sped up the computing time.

Finally, to evaluate our models later, we randomly split our places in half: 50% for training (these act as our queries) and the other 50% for testing (our search catalog).

3. Our Models: BM25 to TF-IDF (and why no Ollama)

To start, we implemented the BM25 algorithm to validate our baseline pipeline. It is a very solid, classic search engine algorithm.

But the goal was to design a model better than BM25. When deciding how to build our system, we thought about using heavier LLMs like with Ollama, especialy after seeing the examples of report you showed in class. However, we thought that for a dataset with tens of thousands of reviews, running a massive generative model locally would be slow and be a bit overkill for this similarity task.

Instead, we decided to build a model based on TF-IDF, paired with cosine similarity. Why this choice? The project instructions actually pointed us in this direction, suggesting TF-IDF as a way to select the best words that describe a place. It is a brilliant method because it penalizes super common words and gives a lot of weight to the rare, specific words that realy define the identity of the places.

4. Evaluation: The Moment of Truth

To judge our models, we used the Ranking Error metric. It is easy to understand. The system ranks all its recommendations. If the very first place retrieved is a perfect match, the error is 0. If we have to go down to the 7th position to find a good place, the error is 6 (n-1). If no place with the same metadata exists in the test set, the error is undefined and skipped. And just to be absolutely clear: the official metadata was only used to evaluate the model at the end, it was never used as an input to design our system.

- **Level 1 Evaluation:** Here, we just checked if the retrieved place was of the same broad type (typeR: hotel, restaurant, or attraction).
    - Our average error with BM25: **0.61**
    - Our average error with TF-IDF: **0.80**
    - *Note:* BM25 proved to be an incredibly strong baseline for broad keyword matching, slightly edging out our TF-IDF model on this simple level.
- **Level 2 Evaluation:** This level was much stricter. To count as a match, the places had to share a very specific detail. For hotels, we checked the price range. For restaurants, we looked at the cuisine and restaurant type. For attractions, we checked the sub-categories. Since some of these are lists, finding at least one matching category was enough to call it a positive match.
    - Our average Level 2 error with TF-IDF: **4.14**.

Conclusion

We thought that getting an average error of 4.14 on the strict Level 2 evaluation is a great result. It means that just by reading some English reviews, our TF-IDF model is usually able to find an incredibly similar place (with the same price range, same food type, etc...) around the 5th position out of hundreds of possibilities. This proves our initial hypothesis: the words people use in their reviews are more than enough to capture the true vibe of an experience.